

Tenet: Agent Memory as a Self-Consistent World Model

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<https://github.com/Nas01010101/tenet>

Abstract

Long-term memory for LLM agents is almost universally implemented as *retrieval over a growing log of past turns*—a document-retrieval abstraction. We argue this is the wrong abstraction for an agent that must model a changing world. We introduce **knowledge churn**—the repeated updating of a fact over a long interaction—and show that retrieval-augmented memory (RAG-memory) *silently degrades* under it: as the number of stale versions of a fact exceeds the retrieval budget k , the reader is handed conflicting values and answers incorrectly. On a controlled benchmark, a strong RAG-memory falls from 100% to 50% current-value accuracy as a fact is updated $2 \rightarrow 12$ times. We propose **Tenet**, which reframes memory as a **self-consistent belief state**—a compact *world model of the user*—rather than a document store. Tenet (i) distills raw turns into atomic, keyed facts; (ii) maintains a **bi-temporal** record so a changed fact *supersedes* its predecessor (retired to history, not overwritten); (iii) enforces **belief-evidence consistency** by retiring raw evidence that echoes a superseded belief; (iv) applies a **predictive-coding write policy**—surprise-gating—that stores only observations the model cannot already predict; and (v) closes the accuracy gap to raw retrieval with **belief-anchored evidence expansion**—spending spare context on query-relevant turns from the sessions the belief state already surfaced. Tenet holds 100% **current-value accuracy across all churn levels** (where a strong RAG-memory falls to 50%), matches strong RAG on retrieval recall (95–97.5%), and—with expansion—**matches its one-shot answer accuracy at equal-or-lower token budget** (57.5% vs 57.5%, gpt-4o reader) while retaining a high-efficiency point at **half the context** and the **best accuracy-per-token** of the systems we evaluate. Tenet thus traces an accuracy-efficiency *frontier* that meets RAG at its budget and beats it at every lower one. We release all code.

1 Introduction

An agent that talks to a user for months does not need a transcript; it needs a *model of the user* that stays true as the user changes. Yet the dominant memory design—retrieval-augmented generation over stored turns [1, 2]—treats memory as a document index: embed every turn, retrieve the top- k most similar, let the reader sort it out. This works for one-shot recall of a *static* fact. It fails, quietly, when facts **change**.

We formalize this as **knowledge churn**. A user who moves cities several times generates many “I moved to X ” turns; all are similar to “where do I live?”, so the top- k fills with *stale* versions. Once the number of updates exceeds k , the latest value may not be retrieved at all, and even when it is, the reader must infer recency from contradictory statements. Accuracy collapses (§4.2).

The root problem is abstraction. A document store has no notion that “I live in Boston” and “I live in Seattle” are the *same fact* with a *changed value*. We argue memory should be a **belief state**: a

compact set of *current* beliefs, each with a temporal extent, updated by observation, kept internally consistent, and queryable across time—the stance world-model and predictive-coding accounts take toward perception [10], brought to agent memory.

Contributions.

1. We identify and name **knowledge churn**, a failure mode of retrieval memory, and give a controlled benchmark exhibiting it (RAG: 100% $\rightarrow$ 50%; §4.2).
2. We present **Tenet**, a memory that is a self-consistent belief state: bi-temporal supersession, a **belief-evidence consistency** rule, and a **surprise-gated** (predictive-coding) write policy.
3. We evaluate on LongMemEval_S and controlled tests: Tenet is **churn-robust** (100%), on par with RAG on recall (95%), **best on accuracy-per-token**, and—with belief-anchored evidence expansion—**at parity with strong RAG on one-shot accuracy at equal token budget**, closing a gap belief-only compression left open.
4. On the standardized **MemoryAgentBench FactConsolidation** benchmark, ingestion-time supersession with zero-LLM deterministic keys **exceeds the published single-hop SOTA at the gpt-4o-mini tier (86.5 vs 78.0 pooled) and ties the multi-hop tier (30.2)**—using only a local 7B backbone, where the benchmark’s 22 systems score ≤ 60 / ≤ 7 .

2 Related Work

Retrieval memory. Mem0 [1] distills salient facts at write time over a vector store; it attaches only a *creation* timestamp and *removed* its graph variant after finding it $3\times$ slower / $2\times$ tokens for a thin gain—evidence we weigh in choosing a light substrate. LongMemEval [2] is the standard long-horizon benchmark; its V2 adds a *latency-aware* metric, a field shift toward accuracy *per cost* that our per-token results target. **Temporal knowledge graphs.** Zep/Graphiti [3] maintain a bi-temporal graph with automatic invalidation—but pay heavy per-write extraction and require graph infrastructure; Tenet keeps the bi-temporal semantics without the graph. **The 2026 bi-temporal convergence.** Concurrent systems adopted bi-temporal supersession: Mem-Strata [7] applies deterministic (s, r, o) supersession over a bi-temporal ledger with no LLM on the read path—and shows *similarity-threshold* supersession leaks stale values where deterministic keying does not, independently corroborating our keyed design; Engram [8] pairs a bi-temporal graph with a hybrid facts-plus-chunks read path (converging on our dual-pool finding); TOKI [9] formalizes contradiction resolution as a bitemporal operator algebra; and [6] sets the current MemoryAgentBench FactConsolidation SOTA with deterministic post-retrieval aggregation. Tenet differs in *where* consistency is enforced—at ingestion and at recall (belief-evidence consistency)—and in coupling the belief state to surprise-gated writes and budget-bounded evidence expansion, evaluated on the knowledge-churn axis none of them report. **OS-style / observational memory.** MemGPT/Letta [4] page memory between context and archival storage; observational schemes keep a stable summary. Both are largely append-oriented and do not treat supersession or forgetting as first-class. **What is missing.** No prior system combines bi-temporal supersession, *belief-evidence consistency*, *predictive-coding write-gating*, principled forgetting, and *belief-anchored evidence expansion* in a light, graph-free store—nor reports the knowledge-churn regime.

3 Method

Tenet stores two layers over one bi-temporal table: a **belief layer** of distilled, keyed facts, and an **evidence layer** of raw turns. Reads never call an LLM.

Distillation. Each turn is distilled into atomic facts, each with a stable semantic key $\kappa = \text{subject:attribute}$ (e.g. `user:residence`), a salience $s \in [0, 1]$, and an event time. The key makes supersession reliable: embedding similarity cannot separate a *restated* fact from a *value-changed* one (we measure a value change at cosine 0.99, indistinguishable from a paraphrase), but a shared key can.

Bi-temporal supersession. Every memory carries event time (`valid_at`, `invalid_at`) and transaction time (`created_at`, `expired_at`). Storing a fact at key κ whose value differs from the current one *supersedes* it: the old fact’s `invalid_at/expired_at` are set; it leaves the current set but stays in history. Current recall filters `expired_at` = $\emptyset$; `recall(as_of=t)` reconstructs the belief set at any past t (time-travel).

Belief–evidence consistency (key rule). The evidence layer answers detail questions but is where stale values hide: “I moved to Boston” survives after the belief moves on. We retire from current recall any raw slice e close to a *superseded* belief:

$$\text{exclude } e \iff \max_{f \in \mathcal{F}_{\text{expired}}} \cos(e, f) \geq \tau_{\text{stale}}, \quad \tau_{\text{stale}} = 0.80.$$

This single rule turns supersession into end-to-end correctness: $55\% \rightarrow 100\%$ (§4.3).

Predictive-coding write policy. A world model stores *prediction error*. On write, a raw observation e is discarded if the store already predicts it:

$$\text{store } e \iff \max_{e' \in \text{store}} \cos(e, e') < g_{\text{surprise}}, \quad g_{\text{surprise}} = 0.97.$$

This bounds the store and drops redundant repetition with no accuracy loss (§4.3).

Forgetting. Rank = relevance $\times$ decay, with $d = 2^{-\Delta t/h} (1 + \beta \log(1 + \text{uses}))$ ($0.6 + 0.8s$), half-life $h = 14$ d; a sweep archives low- d unpinned memories. Retrieval is a *dual pool* (beliefs for consistency, evidence for verbatim detail), each guaranteed a budget share.

Belief-anchored evidence expansion. Compressing a session into a few beliefs wins churn and efficiency but can drop the detail a multi-hop question needs, even when the right session *is* retrieved (recall 95–100%). The top- k result names both the belief state *and the sessions it came from* (each memory’s *source*). Given spare budget B , Tenet fills it with query-relevant raw turns drawn *only from those already-surfaced sessions*—evidence anchored to the belief state, not the whole haystack—under the same stale-echo filter. Setting B to the baseline RAG budget makes expansion never spend more context than flat retrieval; the count m is a knob ($m=0 \rightarrow$ efficiency point; m large under $B \rightarrow$ parity point) that lets one system trace a frontier.

4 Experiments

Protocol. LongMemEval_S (500 questions, ~ 115 k-token histories). We use a **gpt-4o reader** (the same reader Mem0 and Zep report against), a local embedder (**bge-small**), and a cheap **gpt-4o-mini** distiller; the shipped system runs the same code on Qwen Cloud (**text-embedding-v4**, **qwen3.7-plus**) by a config flip. Numbers are compared only to baselines run under identical settings. Baselines: **RAG** (top- k raw turns) and **full-context** (entire history).

4.1 Recall, and the accuracy–efficiency frontier (n=40, gpt-4o reader)

System	mode	recall@10	QA acc	reader tok	acc/1k tok
full-context	—	—	65%*	~ 124 k	0.5*
RAG	top- k turns	95%	57.5%	2,101	27.4
Tenet	efficiency ($m=0$)	97.5%	52.5%	1,067	49.2
Tenet	parity (expansion)	97.5%	57.5%	2,083	27.6

Tenet is a *frontier*, not a point. At its **efficiency** point it answers at half the context (1,067 tok) for the best accuracy-per-token of any system (49.2, $1.6\times$ RAG) at recall parity. Turning up belief-anchored expansion under RAG’s own budget brings raw QA accuracy to **parity with a strong RAG** ($57.5\%=57.5\%$) **at fewer tokens**, closing the one-shot gap belief-only compression left open. Tenet thus meets RAG’s accuracy at its budget and beats it at every lower one; the one category still behind is *multi-session* synthesis (42.9 vs 57.1, up from 28.6), where a question needs several evidence sessions but only some are surfaced. *Full-context, at a weaker reader, spends $100\times$ the tokens for no gain over RAG. Reader-robust: on a cheaper **gpt-4o-mini** reader the parity point edges ahead (Tenet 60.0 vs RAG 55.0); per-token dominance holds across **gpt-4o-mini**, **gpt-4o**, and **claude-opus-4.8** ($\approx 1.6\text{--}1.7\times$).

4.2 Knowledge churn (headline)

One fact updated N times amid distractors, $k = 6, 12$ principals/point:

updates N	2	4	6	8	10	12
RAG	100	100	100	67	58	50
Tenet	100	100	100	100	100	100

RAG degrades monotonically once $N > k$ (-50 pp); Tenet is flat at 100%. Supersession keeps exactly one current value regardless of churn—the property a *belief state* has and a document index cannot. **The curves are identical under a gpt-4o-mini and a gpt-4o reader**: the failure is structural (once $N > k$ the latest value is not reliably retrieved), so a stronger reader cannot rescue RAG.

4.3 Ablations

Belief-evidence consistency. Removing the rule drops Tenet to 55% current-value accuracy with a 45% stale-leak; adding it restores 100% / 0% leak—the single most important mechanism. **Surprise-gating** discards 15% of observations as redundant with no accuracy change, yielding a bounded store where RAG grows unboundedly.

4.4 Standardized conflict resolution: MemoryAgentBench FactConsolidation

On the ICLR 2026 conflict-resolution benchmark [5] (SubEM metric and official reader prompt verbatim; all 800 questions; Wilson 95% CIs), ingestion-time supersession with **fully deterministic, zero-LLM keys** and a deliberately weak **local 7B backbone** scores, pooled over four haystack lengths (6K–262K):

pooled	naive-RAG (same reader)	Tenet	published SOTA (mini / gpt-4o) [6]
FC-SH ($n=400$)	47.8	86.5 [82.8, 89.5]	78.0 / 94.8
FC-MH ($n=400$)	4.5	30.2 [26.0, 34.9]	30.2 / 51.5

Single-hop exceeds the published **gpt-4o-mini-tier SOTA (78.0; our CI excludes it) on a weaker backbone**; every system in the original benchmark table scores ≤ 60 (Zep 7, Mem0 18, MemGPT 28). Multi-hop exactly ties the mini-tier SOTA, where the original table’s best is ≤ 7 . Accuracy barely degrades with haystack length (SH $89 \rightarrow 81$ from 6K $\rightarrow$ 262K): the store is conflict-resolved at ingestion—the property assembly-time aggregation must re-derive at every read. Multi-hop still degrades with length ($42 \rightarrow 20$); reported honestly.

5 Limitations

Multi-session synthesis is the one category where RAG still leads (42.9 vs 57.1). Belief-anchored expansion lifted it from 28.6 but does not close it: these questions need evidence from *several* sessions, and expansion only deepens the sessions the top- k already surfaced—if a required session is not among them, its detail is still missing. Session-diverse retrieval is the natural next step; elsewhere Tenet is at or above RAG. **The frontier is a knob, not free lunch**: parity accuracy costs RAG-equal tokens, and the $1.6\times$ per-token win is at the efficiency point, which trades ~ 5 pp of raw accuracy. **Evaluation**: $n=40$, off-Qwen (gpt-4o / gpt-4o-mini readers, local embedder), one seed; reader stochasticity is $\approx \pm 5\text{--}7$ pp, so the one-shot result is reported as *parity*, not a win. The shipped system uses Qwen Cloud.

6 Conclusion

Treating agent memory as a **self-consistent belief state** rather than a document index makes it robust to the way real knowledge behaves: it changes. Tenet stays correct under knowledge churn where retrieval memory collapses, matches a strong RAG’s one-shot accuracy at equal token budget while offering the best accuracy-per-token we tested, with a light graph-free substrate and no LLM in the read path—and gets time-travel and principled forgetting for free. We hope *knowledge churn* becomes a standard axis for evaluating agent memory.

References

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